Lesson Notes

MODULE 2 | LESSON 1

INTRODUCING STOCKS AND CRYPTOCURRENCIES

Reading Time	75 minutes
Prior Knowledge	Interest, Shorting
Keywords	Stock, Primary Market, Secondary Market, Capital Appreciation, Realized Gain/Loss, Realized, Going Public, Cryptocurrency

In the previous module, we studied fixed income securities, examining the rates and risks of borrowing and lending. In this module, we switch to equities and cryptocurrencies and study not only their expected returns but also their volatilities.

1. Choosing Stocks and Cryptocurrencies

1.1 Stocks

As a bondholder one enjoys certain privileges and one main drawback.

- A bondholder is contractually guaranteed a fixed flow of coupon payments at specified intervals of times. As long as the company does not default, these payments continue until maturity, when the final coupon and principal are repaid together.
- In case of a default, bondholders get paid before equity holders.
- Aside from a situation of default, the amount of the principal returned is equal to the value of the investment made.
- These payments are fixed and if the companies did very well in the market, a bondholder would not receive any gains from that upside. Indeed, this is one of the drawbacks of bonds.

A different type of security, stocks, is aimed at removing the drawback described immediately above. In fact, the ability to participate specifically in a company's upside is the strength of stocks.

Stocks are securities that represent a fractional ownership in a corporation: a **stockholder**, or **shareholder**, actually owns a fraction of the company. Stocks used to be physical certificates that entitled the holder to both a proportional amount of ownership and a share of the company's profits. Today, stock ownership is managed digitally. In the same manner, stock ownership allows the holder to vote on decisions about the management of the company. In reality, most stockholders own such a tiny portion of shares that they have no influence on the management. Collectively, the stockholders tend to vote for a board of directors. In turn, this board of directors selects a CEO and other senior managers, who in turn decide on all staffing decisions among other things.

Another name for stocks is **equities**. This is because stockholders are entitled to the equity ownership of the company. We will see more of this in a later module, but for now, think of equity as the difference between assets and liabilities. This is a simple consequence of the *Fundamental Equation of Accounting*, i.e.:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

from which, using a simple algebraic manipulation, we easily get that

$$\text{Equity} = \text{Assets} - \text{Liabilities},$$

as claimed. From this equation, we see also that the greater the assets, or the smaller the liabilities, the greater the amount of equity the company has. This is effectively owned by the stockholders.

Liabilities and equities are means for a company to raise resources to buy assets needed to produce and/or expand their current lines of business. Companies have two ways to finance their assets: using liabilities (borrowing debts) or using equities via an issuance of shares.

Public companies sell their shares through a stock market exchange (NASDAQ, NYSE, FTSE, etc.). In specific instances, an important managerial decision may have to do with how much debt and how much equity to use to finance a company. In fact, issuing shares could be done as a way to repay back certain liabilities or even most liabilities.

Investors, on the other hand, can buy and sell shares through the stockbrokers. Stock prices are known to fluctuate during the day, but behind their decision to invest, stockholders hope that, over time, the stock they buy goes up in value. Still, this is not guaranteed, and there are instances in which investors see the value of their

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investment in companies go down or, albeit not too common thankfully, even be wiped out when a company files for bankruptcy or goes out of business.

Since investors hope that their investments go up in value, to achieve this goal, they have to choose good companies. Such companies would have some combination of productive assets, well-controlled liabilities, good management, and opportunities to engage in new projects or continue existing projects, creating revenue that exceeds costs. When companies have all these desirable features, they can achieve greater valuations.

The previous paragraph makes it very clear that not all investments in equities are the same, and investors try to build their portfolios combining different strategies that involve different types of companies, or equities if you prefer to use this terminology. One major approach involves selecting certain types of stocks (it is important to become familiar with these terms, as they are frequently used by those working in financial fields):

- **Value stocks** or investments whose value remains stable over time, and in all sorts of market conditions, and may even slowly go up. Investors are attracted to these kinds of companies because they perceive that they are possibly undervalued. Also, these companies are stable and tend to provide a reliable stream of dividends, although that cannot be taken for granted.
- **Growth stocks** are instead investments that exhibit more volatility than value stocks but whose growth outpaces the market average. Investors are attracted to this class of assets as a way to earn from rapid price appreciation instead of relying on current streams of income from dividends.
- **Income stocks** are stocks that offer regular and steady income, usually as dividends with low exposure to risk/volatility. In certain aspects, they are reminiscent of bonds, but ideally, investors buy them to achieve a dividend yield that is above the prevailing 10-year Treasury note coupon rate. In general, income stocks have very limited growth options, although they can earn a small return from the appreciation of the value of shares. Unlike value stocks that are desirable for the possibility of undervalued assets, income stocks are sought after as a chance to bolster fixed income portfolios.
- **Dividend aristocrats** is a name that usually attracts attention, but they are just companies in the S&P 500 that have increased their dividends annually for **the most recent 25 years**. They are not very common since, as of May 2024 (when this lecture was created), there were only 68 stocks in this category out of about 6,000 listed on either the Nasdaq or NYSE. These shares usually have low volatility and allow shareholders to earn in two ways: price appreciation of their shares and dividends appreciation.

Once investors have decided that they want to own a company's stock, how do investors buy shares? There are two ways to buy a stock:

- **Primary Market:** When the firm decides that it wants to issue new shares of stock, you can decide to buy directly at the time of issuance. Purchasing a share from the issuer directly occurs at an **initial public offering** or **IPO**. This is usually difficult to do for small investors. In fact, when a company decides to go public and issue stocks via an IPO, such an IPO is underwritten by one or more investment banks or broker-dealers. They purchase the shares from the company and then try to sell those shares to investors. Broker-dealers try to sell large pieces, and thus, these shares usually end up in the hands of large institutional investors. This is not because the small investors are not important, but the way things happen is faster and more efficient from the point of view of the bankers and broker-dealers involved.
- **Secondary Market:** Once the shares of stock have been issued, they can easily change hands by trading on exchanges. If you wanted to buy existing shares, then you would have to convince an owner to sell their shares at a mutually agreed-upon price. Such convincing simply means buying at the owner's offer price on a stock exchange. Similarly, if you wanted to sell your shares, then you would have to find a buyer willing to pay a price you both agree upon. Indeed, there is an extensive amount of trading on some stocks that have been issued. From the firm's point of view, this is a good thing because the company's shares are seen as marketable and liquid (terms we will formally define later in this course).

As an investor, why do we want to purchase stock? From a corporation's perspective, why do they want to issue stock? Let's tackle that next.

1.2 Stock Benefits

As investors, from our discussion in the previous subsection, we can easily see some of the advantages that come from owning stocks:

- **Capital Appreciation:** Stocks can go up in price over time. In other words, you can "*buy low, sell high.*" Capital appreciation means that the asset you purchased increased in value, so when you sell it, you earn the difference in price as a profit. Therefore, you can buy stocks that you believe are underpriced or undervalued, hold on to them for some period, and sell them once they have achieved that higher value. This strategy is the essence of value trading. However, you may also buy a very popular stock that is already expensive and hold onto it while it increases even more in price. Capital appreciation simply refers to the extra value your asset achieved while it was in your possession. Anyone who's sold a home whose value has risen over time understands the beauty of capital appreciation.
- **Dividend Payments:** When companies make profits, they can decide to do one of two things: first, they can pay stockholders a dividend. This dividend is income the company has made (not necessarily in the latest quarter) that can be distributed in proportion to the number of shares you own. Second, a company can decide to retain earnings they will subsequently use for funding existing or new projects. Such projects could range from building new infrastructure that will eventually lead to greater earning potential or even acquiring other companies that enhance shareholder value long term.

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- **Voting Rights:** As owners of the company, stockholders are endowed with rights to vote on management issues. It's somewhat of a democracy in that the more shares you own, the more votes you get. However, it's also a representative democracy in that these votes are cast for members on the board of directors. This board of directors in turn tends to vote for the executives, which in turn select and hire workers. Stockholders can have some voice but generally not over the daily operations.
- **Ease of Trading:** Owning a stock means that you have an asset that is publicly traded in an accessible exchange. As a result, its price is clearly known and can be traded at or very close to that value for the several hours a day that the exchange is open. These exchanges are typically open five days a week during business hours. Therefore, it is easy to convert the stock to its cash equivalent if you decide to sell or to simply buy the stock at a fair price when you wish to own it. Throughout the day, you know how much your holdings are worth. Other assets for which no exchange exists, such as art or real estate, are more difficult to value.
- **Extra Income through Securities Lending:** In the previous module, we discussed shorting. As you recall, shorting involves borrowing a security that you don't own, selling it in the marketplace (which gives you cash), and waiting for capital depreciation to occur. Capital depreciation means that the price drops, after which you would buy back, or **cover**, the short, collect the difference between the high price at which you sold and the low price at which you bought, and then return the security to its original owner. As the owner of a stock, you can be on the lending end of the security. Just like with a loan, whenever a lender is lending a valuable asset (e.g., cash in a loan), the borrower pays the lender income (e.g., interest for a loan). In the case of a stock, the borrower pays the lender the financing cost, which is a type of "daily rent" for the security. As a stock owner, if you wish to have extra income, you can let the broker (who typically holds your security) lend out your securities to short sellers. During this time, you could still sell those shares yourself. While your long position is loaned out, you would be collecting some financing costs, in addition to the dividends that the stock pays. This extra income from financing costs continues until the borrower covers the short.
- **Centralized Depositories:** The record of ownership of the stock is very safe. It would be virtually impossible for there to be a mix-up as to who owns the shares. This safety is part of a robust financial system in which the exchanges, brokerages, and settlement houses keep very careful records of who owns stock and how the stock has been traded. Compare this to real estate. Real estate deeds are typically held at local government (e.g., county or prefecture) levels, of which there can be thousands in a single country. In fact, it is possible for two people to make claim on the same title of land. For instance, after you pay someone for a home and receive the title, someone might come in and make a claim for the same title. This potential discrepancy explains why there is something called **title insurance**, which is insurance that if you purchase a home, you have the exclusive rights to its title. Stocks avoid this confusion by efficiently centralizing many of the processes with the exchange and clearing houses. Note that these benefits are not specific to stocks but are really a benefit of having exchanges as the arena for trading in addition to well-established clearing agencies and custodians for transferring and maintaining ownership records.

We can also look at the benefits of stocks from the point of view of the company that issues them. If a company wishes to grow, they have two primary ways of raising capital:

- **Issue Bonds:** As we know from the previous module, companies that issue bonds are borrowing funds. They can raise any amount of capital that the market is willing to supply. But when that first coupon date arrives, they will have short-term cash-flow obligations. Suppose a company issued a 10-year bond. In six months, they will owe interest on the bond to each bondholder. If the company were unable to pay the interest because those funds were tied up in infrastructure for long-term projects, then they could default as early as the first interest payment. Therefore, companies issuing bonds or taking out loans have to take into consideration the timing and sizes of the obligations due.
- **Issue Stocks**. Instead of borrowing funds, the company can decide to sell stocks. By issuing shares of stock, they are willing to transfer some of the ownership of the company to any members of the public willing to pay the share price for it. At an IPO, the share price is determined by an underwriting bank, which tries to place the shares with investors. Indeed, there is often a long list of willing participants eager to jump into an exciting new venture. Historically, some stocks perform exceptionally well on the day they IPO.

As seen here, the advantage from the company's point of view is that the stock never has to be paid back. It simply cedes an amount of ownership to others. Imagine if our class started a company. Then, all the students (and the professor) would own all the assets and have rights to all future revenues. However, suppose our collective capital were insufficient to invest in the required infrastructure. In that case, we could issue shares of stock that would dilute our percentage of ownership by granting ownership to whichever parties purchase the stock. This is what it means to "go public." For companies that have long stages of growth or require long cycles to extract a return on investment, stock issuance is much safer than bond issuance because it avoids coupon repayments. During this time, there will very likely be no dividend, but willing investors will not expect to receive them while the company is in a growth phase. This idea will be discussed later in this module.

1.3 Disadvantages of Stocks

Each of the benefits of owning stocks that we outlined can certainly be turned around. For example, as easily as capital may appreciate, it may also depreciate, which means you can get less money out of your stock then you put in. Certainly, if you had bought a house at the height of the market and then sold it after the market crashed, you would take a loss. With regards to stocks, such a loss may only occur on paper if you continue to hold the stock. However, if you actually sold the stock at a lower price, then you have **realized** the loss. So unlike other investments, like our certificate of deposit, stocks can and often do lose money.

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Another disadvantage of stocks is that dividend payments may not occur. Even if a company promises to pay dividends, you might receive less than expected or nothing at all. Companies can change their strategy. First of all, a company might not have income to pay out. Second, the company may decide to retain earnings rather than distribute them to shareholders.

While holding stock gives you voting rights, the proportion of your shares is likely to be so small that the effect is negligible on any results. Often, a majority of shares can be held by a few--even a single shareholder--who essentially control the company. Also, there can be a type of investor called an **activist investor.** For example, Bill Ackman is an activist investor. He chooses firms very methodically and buys a significant number of shares, giving him voter power, which he then uses to influence the management to "shape up or ship out": that is, improve their performance or quit. Activist investors can add volatility and uncertainty to a stock by the pressure created between shareholders and management.

In case of bankruptcy, equity holders are the last in line to be paid, and they will be paid only if there is any money left after paying the other categories of investors. Secured creditors are the first in line to be paid, possibly after the company has been liquidated. The group of secured creditors includes bondholders. Then, unsecured creditors are prioritized in a certain order: company employees, tax agencies, possible environmental claims, and, at the very end of the list, preferred and common stockholders.

Additionally, it's important to note that the market is not 24/7. Stocks do not trade 24 hours a day or 7 days a week. Suppose there is bad news when the market is closed. The company may lose value, so you want to sell your shares. However, you might not be able to do so until the next business day, which could be one, two, or even three days later, depending on weekends and observed holidays. As a result, holding stocks can be risky if there is any news that can greatly affect the stock price.

Just as there are disadvantages to the investor, there are disadvantages to the company. From the company's perspective, issuing stock means that the company can lose a lot of control over operations. When you look at companies that were founded by visionaries of their times, such as Ford Motor Company founded by Henry Ford, they have been able to keep a good amount of stock in the family. As of a few years ago, the Ford family held a significant amount of the voting shares of stocks, which gave them 40% of the voting power over the board of directors. On the other hand, suppose Ford sold so many shares, say to an activist investor, that the Ford family members lose control of the company. This new activist investor can vote in a new board of directors, which can create poor leadership and management. Issuing stock can disrupt the ownership, and consequently the management, of a company. Issuing bonds avoids that problem and would have been safer in that case!

1.4 How to Value a Stock

We mentioned that one of the reasons behind investing in stocks is capital appreciation. To assess a stock's potential for capital appreciation, investors must have a reasonable estimate of the value of a stock or need to get such value from stock analysts. While quantitative analysts may not get particularly involved in this, some knowledge is nevertheless useful. There are several techniques that can be used and they are all more or less rooted in models of corporate finance. One of these techniques, the **Discounted Cash Flow model** (DCF), is very general and requires a deep knowledge of how the company works. This type of modeling is out of reach for someone who does not know the company very well. Other models like relative valuation or the **dividend discount models** (DDM) are much more accessible. In particular, the Gordon Growth Model is relatively simple yet reasonably powerful and does not require specialized knowledge of the business of a company.

1.4.1. Gordon Growth Model (GGM).

This special dividend discount model is based on the idea that the stock's present-day price is worth the sum of all its future dividends when discounted back to its present value. The price so calculated is called **fair value**. Then, if the current price of the stock is lower than its GGM value, the stock is undervalued and we should buy it. On the other hand, if the GGM value is below its current price, then we should sell the asset or maybe even short it. This model is based on the time value of money, which we are already familiar with and requires only 2 key inputs:

- **Estimating dividends and dividend growth rate**. Estimating the value of a company's dividends going forward is not easy in general, but we can make certain assumptions. For instance, we could extrapolate a trend in past dividend payments and use that to estimate future dividends. In particular, we can assume that the company has a fixed growth rate of dividends and that this flow of dividends continues unchanged forever (the technical term is "in perpetuity"). So we need the expected next closest dividend per share in time, which we denote by D_1 and the growth rate for dividends, which we call g.
- **Estimating a discount factor**. Shareholders who invest money in stocks are aware that they are making a risky investment, and as with all risks, they want to be compensated for it. The firm cost of equity capital is usually taken to represent what the market demands as compensation for owning the asset and assuming the underlying risk. This cost of equity/compensation rate will be denoted by k. Note that k has to be greater than g so that the denominator remains positive.

The formula for the GGM model is the following

$$\text{Value of stocks} := P_0 = \frac{D_1}{k-g}$$

Example. Suppose that company ABC is expected to pay a dividend of $1.5 per share, the expected dividend growth rate is assumed to be 2%, and its cost of equity capital is 8%. What is the DDM value of ABC stock? What

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should an investor do if the current price of ABC share is $30?

Answer. In this case, $D_1 = 1.5, g = 0.02, k = 0.08,$ and the DDM formula yields

$$\text{Value of ABC stock} = \frac{1.5}{0.08 - 0.02} = 25.$$

As the DDM fair value is $25/share and ABC is currently trading at $30/share, the company is overvalued and the investor should sell their shares of ABC.

Example. *A variation of the problem above consists in saying that ABC just paid a dividend of $1.4, and that its dividends are expected to grow at 2.2% a year in perpetuity. Its cost of equity is 8%. What is the DDM fair value of ABC shares?*

Answer. Here, the student must pay close attention. In this case, we are not given the expected dividend (D_1), but the most dividend paid (usually denoted by D_0). We need D_1 in our formula. This is not a problem because

$$D_1 = D_0 \times (1 + g) = \$1.4 \times (1 + 2.2\%) = \$1.43.$$

Then

$$\text{Value of ABC stock} = \frac{1.43}{0.08 - 0.022} = 24.66.$$

The GGM model is easy to use (and it requires only 3 inputs) but has some disadvantages:

- It can only be applied if a company pays dividends and this makes it suitable to value stable companies with well-established dividend cash flows. However, the method would be very unreliable or even completely inapplicable in the case of a start-up company.
- The constant dividend growth rate is also applicable mostly to mature, stable companies and difficult to justify for growth assets for which the growth rate is likely much more volatile.
- It assumes that the company will exist forever, be able to pay dividends forever, and even be able to grow them forever at a given rate.
- From a mathematical point of view, we need the cost of capital to be greater than the dividend growth rate.
- External market conditions are ignored.

1.4.2. Multistage Gordon Growth Model (mGGM).

This model maintains most of the assumptions as GGM but allows for a period of fast growth and a period of stable growth. The idea is that companies may grow fast during early stages and then they become mature and grow at a lower and more stable rate. To use this model, we need the following inputs:

- Expected dividend (D_1), or most recent dividend paid (D_0);
- Fast growth rate (g_H);
- Number of years the company remains in a high-growth period (n);
- Growth rate in the stable period (g_L); and
- Cost of equity capital (k).

Then, the formula for P_0 is more complex, but nothing too scary:

$$P_0 = \sum_{i=1}^{n} \frac{D_0(1+g_H)^i}{(1+k)^i} + \frac{D_0(1+g_H)^n(1+g_L)}{(1+k)^n(k-g_L)}.$$

The formula may look a little daunting, but it can easily be implemented via a spreadsheet. Note that this model can be extended from 3 to a number N of stages, if necessary.

1.4.3. Multistage Gordon Growth Model with Linearly Declining Growth Rate (H model).

This is essentially the same model as the mGGM, but it allows for the growth rate to decline linearly from high growth to stable growth over a period of H years consisting of three stages so that the assumptions are now:

- Stage 1: High growth rate is g_H; stage 1 consists of only 1 period.
- Stage 2: The growth rate transitions from g_H to g_L over a period of H periods.
- Stage 3: From time $H + 1$ on, the company grows in perpetuity at the stable growth rate of g_L.

The formula in this case actually allows for a closed-form solution:

$$P_0 = \frac{D_0(1+g_L)}{k-g_L} + \frac{D_0(H/2)(g_H-g_L)}{k-g_L} = \frac{D_0(1+g_L) + D_0(H/2)(g_H-g_L)}{k-g_L} = \frac{D_0}{k-g_L} \times [(1+g_L) + (H/2)(g_H-g_L)].$$

Example. *Suppose we are given the following information about company STU*:

- *current stock price = $34*
- *cost of equity = 9.5%*
- *last dividend paid $1.85*
- *growth rate (stage 1) = 7%*

- *growth rate (stage 3) = 3%*
- *years (stage 2) of linear decline = 10*

What is the fair value of company STU? Is it overvalued, undervalued or fairly valued?

Answer. Using the following spreadsheet, we find $P_0 =$ 35.01. $Hence, thesharesof$STU $ are mildly undervalued.

1.4.4. True Multistage H Model.

The previous model was important to illustrate the technicalities of the H model. Now, we extend it to the following:

- Stage 1: High growth rate is g_H; stage 1 consists of m periods.
- Stage 2: The growth rate transitions from g_H to g_L over a period of $H = n - m$ periods.
- Stage 3: From time $n + 1$ on, the company grows in perpetuity at the stable growth rate of g_L.

We could provide a formula for this case too, but it is going to become cumbersome. Therefore, we will utilize a spreadsheet and show how to apply it via an example.

Example. *Company PQR is currently trading at $27.14 per share and just paid a dividend of $0.55. We also believe that PQR can grow at 10% for the next 5 years, and then, its growth rate will decline to 5% linearly over a period of 10 years. Using the CAPM model, we also determine that its required return on equity is 8.5%. What is the fair value per share of company PQR?*

Answer. This is contained in the following spreadsheet and we can see that the value of PQR is $24.75 per share; in other words, the current market value of PQR is overvalued by about 9.6%. You can see from the spreadsheet how the previous model can be used to work for a true multistage H model.

1.5 Stocks vs. Bonds

Given how important these topics are, let's turn to stocks and bonds and further compare them. When a corporation issues a bond, the bond buyer is effectively providing a loan to the company with the repayment date of the maturity date. During the life of the bond, the bondholder receives periodic interest payments. When a corporation issues stock, the stock buyer is effectively giving a loan to a company as well but with no repayment date. That is, the money collected when stocks are issued is never paid back by the company to the stockholder. Instead, the stock buyer is effectively getting ownership of the company directly.

That is the main conceptual difference between the bond buyer and the stock buyer. While both types of investments are providing funding to a company, the bond buyer is doing so conservatively. Their funds will almost assuredly be paid back, so long as the company has sufficient assets. For the stockholder, there is no guarantee that they will receive anything. Typically, the stock price will not go to zero, but it can be much, much lower than when it was traded. For this reason, bonds are safer than stocks. This explains why bondholders are paid first and paid fixed amounts at specified times, all before any money can be distributed to stockholders.

For the bond buyer to receive interest and principal, the company merely needs to remain solvent. For a stockholder to receive dividends, the company must remain solvent, be profitable, and decide to pay dividends (as opposed to retaining the earnings). Since that is more difficult and uncertain, stocks tend to be more volatile. This causes stock prices to fluctuate more wildly than bond prices do.

Finally, for a stockholder to receive capital gains, the company has to create the perception (even if it is not real) that there is greater value, which justifies a higher share price. Of course, companies might fail to do this, which creates more risk. The lack of guarantees on dividends or price gains makes stocks riskier.

Example. *Suppose Cindy bought 100 shares of company ABC at a price of $12.44 each. During the 1-year period Cindy holds the stock, ABC pays 4 dividends of $0.60 each. Then, immediately after receiving the fourth dividend, Cindy decides to sell the 100 shares and she is able to do so at $15.13 per share. Compute the rate of return for Cindy. For simplicity, assume that Cindy is not re-investing the dividends as she receives them. Let's say she spends them immediately.*

Answer. Let's use a scheme that shows the dates and the cash flow.

- time $T = 0$: Cindy has a negative cash flow of 100*$12.44 = $1244 to buy the 100 shares of company ABC.
- time $T = 1/4,$ she receives the first dividend, i.e., $100 *$ 0.60 = $60. $
- time $T = 1/2,$ she receives a second dividend of $60.
- time $T = 3/4,$ she receives another dividend of $60.
- time $T = 1,$ she receives the fourth dividend of $60 and she also receives 100*$15.13 =$1513 from the sale of her 100 shares of ABC (we ignore commission and fees).

Computing the return over the 1-year period is very simple, and it amounts to

$$\frac{60 \times 4 + 1513 - 1244}{1244} = \frac{509}{1244} = 40.92\%,$$

a sizable return prompted by a generous dividend yield of 0.60 X 4/12.44 = 19.29%. and a respectable capital appreciation of 1513/1244-1 = 21.62%.

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2. Cryptocurrency

Many students who take this course have only a vague notion of cryptocurrencies. Hence, a quick introduction is beneficial. Cryptocurrencies are based on **blockchain technology**. As a technique, blockchain goes back to the early 1990s, more precisely to the work of Haber and Stornetta in 1991. Then, in 2002, Back introduced the idea of **proof of work**, and finally, in 2008, Satoshi Nakamoto used these ideas to introduce Bitcoin.

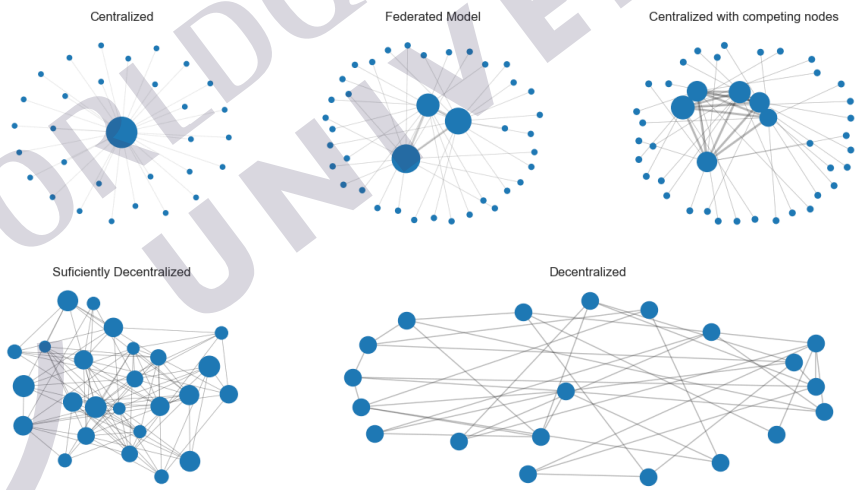
We surely do not have the time and space for an accurate treatment of cryptocurrencies, which would require hundreds of pages, but it is important to understand the main ideas behind the concept.

Those who are familiar with accounting are certainly aware of what a **ledger** is. Nevertheless, we give the definition here as well for the benefit of those students who have never been exposed to accounting. A ledger is a book, possibly a digital book, that contains bookkeeping entries, and it is used to summarize information for all of a company's financial transactions over a pre-specified period. A ledger then serves as a master document.

A blockchain is a **shared** and **immutable** ledger that facilitates the process of recording transactions and makes it possible to track assets in a business. A business relies on this information to function, and the faster and more accurate the information is, the better. In its essence, a blockchain network can track orders, payments, etc., and all members within the company share a single view of the actual business, can see all the details, have confidence in what the company is doing, know what the company does well or not so well, and gain a sense of what opportunities exist. We could say that a blockchain is a **distributed ledger technology** or a **distributed database**.

Traditionally, the database/ledger is entrusted and controlled by a centralized figure or organization. When Nakamoto introduced Bitcoin, he created a single distributed database that is accessible to everyone, i.e., anyone is able to see the balances and enter and submit new transactions at any time, but the ledger is not controlled by anyone specifically and there is no need for permission to access or use it. In this sense, the ledger is maintained on a decentralized basis. In centralized organizations, different parties send their information to a central database, then someone checks that everything is submitted properly before the information is entered into the official database. On the other hand, in a decentralized situation without a central control center, if everyone involved can agree on the status of the database at any time, all the databases can be synced, thereby saving time and usually reducing cost as well.

Figure 1: Decentralization Is a Spectrum



The figure illustrates a possible path of decentralization in the context of blockchain networks. The three upper graphs depict permissioned centralized networks, where large circles represent nodes and small ones represent users. The size of the nodes represents their significance within the network. As decentralization progresses, the significance of each node becomes more uniform. The lower two graphs represent permissionless and decentralized networks, with the lower right graph (Decentralized) being more conceptual than practical.

Clearly, if the idea of a permissionless ledger is to exist and function, there are a number of technical challenges that need to be solved. Think of this: essentially, in a decentralized world, we have several copies of the same database floating around on thousands or millions of different machines and, since no one is in charge, we have to make sure that all copies are identical, are updated together, and reflect real transactions (as opposed to fraudulent ones). The solution to these problems is what gave us blockchains:

- All participants (nodes) have a copy of the same ledger;
- Once a ledger is accepted as the true ledger, it cannot be changed anymore and all transactions that are validated are irreversible;
- Transactions must be time-stamped to avoid the possibility of adding the same transaction several times;
- All nodes agree on the validity of each block and the blockchain as the whole of the individual blocks;
- The participants remain anonymous;

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- All transactions/records are encrypted; and
- Blockchains are programmable. This is a bit difficult to understand without any previous knowledge of the field, but it is a necessary feature to be able to use smart contracts.

Is a blockchain secure? The question is legitimate because, without trust, people would never share valuable information with people they do not know and/or do not trust. Blockchains store data using sophisticated math and complex innovative software rules that are difficult for an attacker to penetrate and modify. However, this does not mean that they are completely unbreakable.

Bitcoin's blockchain contains the shared history of every Bitcoin transaction ever made just like an accounting ledger. Multiple copies of this ledger are stored on a network of computers and these computers are called **nodes**. Each time someone submits a transaction to the ledger, the computers/nodes check that the transaction is valid (for instance, they check that whoever used bitcoins to pay for the transaction actually had those bitcoins to spend). The valid transactions become blocks, and the blocks are then added to a chain that already contains previous blocks. These nodes have owners, and the owners of these nodes are called **miners**. When miners successfully add new blocks to a chain, they earn a reward (bitcoins). Thus, we now have a sense of how a cryptocurrency like Bitcoin is created.

Now that we understand what a block is, we can describe the forms of protection that make the blockchain secure:

- each block has a cryptographic **fingerprint** (called *hash*);
- a **consensus protocol** that is the process by which the nodes in a network agree on a shared history.

Initially, the fingerprint requires a lot of computing time and energy to generate. It is used as proof that the miner who added the block to the blockchain did the computational work in order to earn the bitcoin reward (**proof-of-work protocol**). Then, the hash becomes a seal, and altering the block would require generating a new hash. However, verifying that a hash matches its block is easy. Once the nodes have verified the hash, they update their copies of the blockchain by adding the new block. This is what we referred to earlier as the consensus protocol. Finally, the hashes also serve as the links in the blockchain in the sense that each block includes the previous block's unique hash. This is important because if we wanted to change an entry in the ledger later, we would have to calculate a new hash for that block and for all the blocks that come after it. And there's more: we would have to do it faster than the other blocks can add new blocks to the chain, which requires computers that are more powerful than those used at the other nodes of the network combined! This gives us a chance, but not a guarantee, of being able to pull off that trick; otherwise, there would be conflicts among some of the nodes, and the alterations would be rejected.

We also mentioned smart contracts. Ethereum, a decentralized blockchain introduced in 2015 by Vitalik Buterin, is an example of a platform that allows for the running of computer programs called smart contracts. Hence, the code of the programs can create and transform data or tokens on top of the blockchain of which it is part. Smart contracts control assets and data and define transactions between assets, data, and network participants. In this sense, they are quite powerful, as a user could encode rules to define any type of transaction, which could create new and scarce assets for special purposes. Ethereum allows anyone to deploy permanent and immutable applications on it. This can be seen as one of the reasons for decentralized finance (DeFi) to exist. Furthermore, on Ethereum, users are allowed to interact not just with other users but even with smart contracts. Ethereum uses **Ether** as the native currency for the platform. We can't get too deep into this topic, but smart contracts, called ERC (*Ethereum Request for Comment*), can establish certain rules. For instance, ERC-20 is a fungible token, and we could use this smart contract to establish that a 1 EUR bill has the same value of another 1 EUR bill. ERC-721 is a smart contract that establishes that each token is associated with a particular asset and thus not fungible. While Bitcoin uses a proof-of-work consensus mechanism, Ethereum uses a **proof-of-stake mechanism** instead.

The software applications we mentioned earlier live on a decentralized smart contract platform. For this reason, they are called **decentralized applications** and they are **permissionless**, i.e., anyone can use them and there is no central body that controls them. Instead, there is a collectively owned organization called DAO (**Decentralized Autonomous Organization**) with its rules of operation encoded in smart contracts and determining who can execute what behavior to upgrade. Ethereum **wallets** are applications that give individuals control over their accounts. They contain everything one needs to prove one's identity and they can handle one's assets. Individuals can sign in, check their balances, send transactions, etc. Essentially, wallets are tools for interacting with Ethereum accounts. Any user with any amount of ETH can help make the networks secure and earn rewards in the process in exchange for running software that properly batches transactions into new blocks and checks the work of other validators.

The Ethereum blockchain is only aware of what happens on the Ethereum blockchain. But what if the information needed is outside of such a blockchain? This is where an oracle solves the problem. An oracle is a bridge between the blockchain and the external world, providing a way to bring off-chain data onto the blockchain. Oracles are especially important since many meaningful applications need to use data from around the world. Let's assume, for example, that we have a decentralized application that accepts bets on a soccer match. We have three possible outcomes: Team A wins, Team B wins, and a draw. Let's also assume that this application has gathered a lot of bets that will be distributed among those who selected the correct outcome. The question now is who is responsible for inputting the correct result into the app? It is easy to understand that malicious actors have every motivation to find ways to input the data they want into a decentralized application. To solve this problem, blockchain engineers have created various types of oracles such as centralized, decentralized,

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consensus-based, and human-based. The choice of each type depends on the tradeoff between safety, ease of use, simplicity, and cost of operation.

In our quick introduction, we talked about Bitcoin and Ethereum, albeit briefly, to ensure that every student has at least a modicum of understanding of cryptocurrencies. But there are many more besides Bitcoin (BTC) and Ether (ETH): Tether, Solana, Litecoin, Bitcoin Cash, BNB, Dogecoin, and many, many more. Then, there are online exchanges, for instance Coinbase, where one can buy major cryptocurrencies, and you can use bitcoins to buy a computer or ethers to play digital games, etc.

Some people look at cryptocurrencies as investments, and they behave like other investments. For instance, it is not unusual for BTC to increase or decrease in by price by $\pm 10\%$ on any given day. Smaller cryptocurrencies can be even more volatile.

Shareholders may find the selection of stocks daunting considering the bewildering number of stock choices in different countries on different exchanges. And, even if we selected a specific company, we may find that its current stock price is too high. So while we did our homework, it's possible that the opportunity to purchase at a good price is long gone. In this case, we could decide to short the stock. However, short selling is very risky because the potential loss is unlimited. The price can increase without bound, causing the loss to increase without bound! Hence, we could decide that we want a different asset class all together, and this is where cryptocurrencies could become appealing to investors.

Cryptocurrencies are digital currencies that have disrupted the financial system. Typically, money was issued by a government, perhaps backed by gold, and served as an exclusive form of legal tender for a specific country or multinational economic association (e.g., the euro in the European Union). Cryptocurrencies transcend borders. They lack central regulating or issuing authorities. Bitcoin originated just as the world was experiencing the Great Recession. Trust in Bitcoin has replaced, to a degree, trust in the prudent, unbiased decision-making authority of governmental and central banks with “objective, rational, incorruptible mathematical calculations” in a public decentralized ledger (Stellinga et al. 7). Indeed, the advantage of cryptocurrencies is that they are using a decentralized system to record the entire history of transactions.

2.1 Cryptocurrency Benefits

Now that we have provided an overview of what cryptocurrency is, let's discuss its benefits. First, cryptocurrencies have extreme returns. You may find a stock that trades aggressively and whose price fluctuates, but it would be much easier to find a cryptocurrency that trades with greater amounts of volatility. If you believe that you have a good way to time markets (you know how to buy low and sell high), then cryptocurrencies might be right for you.

Another advantage is that there is exponential growth. Less than twenty years ago, cryptocurrencies didn't exist. Now, there are at least 10,000 active cryptocurrencies. Today, there are between $1 and $2 trillion (two million million) dollars in several different cryptocurrencies.

The third benefit is that cryptocurrencies trade around the clock: 24 hours a day, 7 days a week. This is because current cryptocurrencies defy borders; that is, they do not physically exist in a location like the Tokyo Stock Exchange that closes during off-business hours in Japan. Human traders go home after a day's work, but the cryptocurrency markets are always open! The ability to trade at any hour means that you can react immediately to news. If there is significant news that impacts the market, you can trade quickly without waiting until the next business day. This can be a big advantage when compared to stocks. For example, suppose you had sold short Peloton stock going into a weekend. Over the weekend, it was announced that there might be a takeover. By the time the stocks opened the following Monday, the stocks increased over 30%. If you were a short seller, you would have lost 30%. Around-the-clock market access would have prevented that, but it is not a feature of stocks like it is for cryptocurrencies.

2.2 Cryptocurrency Disadvantages

There are most certainly disadvantages to cryptocurrencies. It is very difficult to apply the value mechanism that we use for stocks to cryptocurrencies. This is because cryptocurrencies tend not to pay dividends. Therefore, it is hard to understand the value. Indeed, some investors have shunned Bitcoin. One of the world’s most famous investors Warren Buffett said that Bitcoin is “probably rat poison squared” ("Crypto is 'rat poison'") and referred to it as a gambling instrument. In 2021, famous financial practitioner and author Nassim Nicholas Taleb said that Bitcoin is worth zero and fails a hedge (Macheel). JP Morgan CEO Jamie Dimon was quoted as saying that “Cryptocurrency has no intrinsic value…You are basically buying a token” (Helms).

The only way to make money in cryptocurrency is to hold it until some other investor believes it's worth even more. Unlike stocks, which are issued by companies that make products or provide services, cryptocurrencies are an implied value that's recognized by the holders at large. Be sure to distinguish cryptocurrencies from real-world currencies. Real-world currencies are known as foreign exchange or FX. This is where you could trade, for example, using the exchange rate between Great British pound sterling and Japanese yen.

2.3 Bitcoin vs. Bonds

Finally, let's discuss the difference between Bitcoin and bonds. Indeed, these are probably the two ends of the spectrum in terms of return. Risk-free bonds are likely the most conservative investment, in that you are virtually guaranteed your principal back plus a specified amount of interest so long as the issue remains solvent. Bitcoin, on the other hand, has little if any intrinsic value. Again, we emphasize that cryptocurrencies are not from companies that make and issue products and services. Cryptocurrencies are adding value in that they are

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providing a publicly distributed ledger for a huge number of applications both in and outside of finance. Nevertheless, the value is quite subjective. That being said, Bitcoin has gone from zero to tens of thousands of dollars. Even as these lesson notes are being written, Bitcoin is projected to achieve almost $100,000 per Bitcoin. It would take a stock investment many years to achieve such a high increase, but if you held a bond to maturity, you are effectively earning the yield to maturity. If you paid par for the bond, the yield to maturity would equal the coupon rate. This yield, or interest rate, is a small fraction of the overall principal, so the overall return is not very large. However, in Bitcoin, it is possible to make 50%, 100%, 200%, or even 1,000% or several thousand percent. Keep in mind, though, that what goes up in markets can also come down. It is just as easy if not easier to lose 90% of your money. Consequently, it's possible you could have bought Bitcoin and experienced a huge market correction of 40 or 50%. You can see stories where athletes are paid in Bitcoin. For some, they find that by the time they go to use the funds, they have a considerably smaller amount to withdraw from ("Buy Bitcoin Worldwide").

3. Conclusion

In this lesson, we looked at the basics of stocks and cryptocurrencies, highlighting their advantages and disadvantages. Compared to fixed-income securities, stocks and Bitcoin have more upside potential but also more downside risk. Choosing the right one at the right time is key. In the next lesson, we will examine the factors to consider when selecting stocks and cryptocurrencies.

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